HAT-P-3b

R-1365

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_HATP-3_180403` · created 2026-08-01T00:04:03+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458211.697 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1178 +/- 0.0086
Transit depth (Rp/Rs)^2	1.39 +/- 0.2 [%]
Orbital Inclination (inc)	87.18 +/- 0.49
Airmass coefficient 1 (a1)	0.9889 +/- 0.0091
Airmass coefficient 2 (a2)	0.0084 +/- 0.0029
Scatter in the residuals of the lightcurve fit is	0.92 %
Best Comparison Star	#1 - [497, 110]
Optimal Aperture	4.66
Optimal Annulus	14.02
Transit Duration (day)	0.0882 +/- 0.0043

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1178 ± 0.0086 vs 0.1063 (deviation 1.04σ)
Duration fit vs published	0.0882 d vs 0.0867 d (deviation 0.28σ)
Mid-time O–C	-2.6 min
Depth SNR	10.7
Residual scatter	0.92 %

Light curve

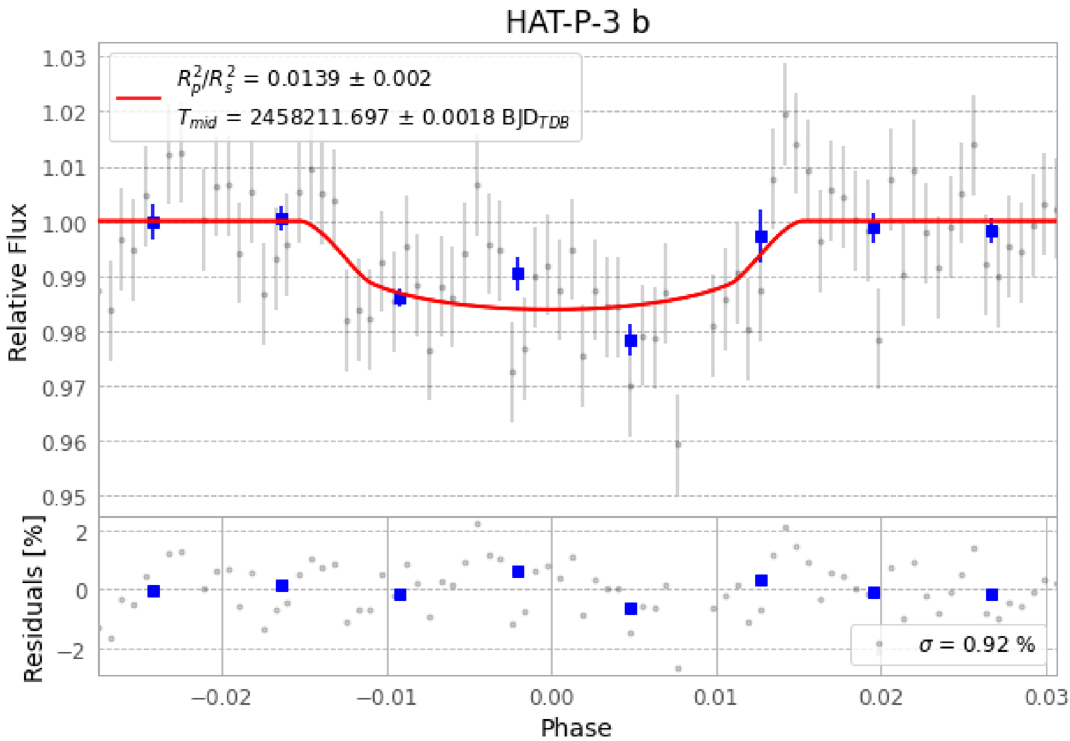

Figure 1 — FinalLightCurve_HAT-P-3 b_2018-04-02.png (SHA-256 210f86703ae2e8f8…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [417, 189], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 1bcea36f8f827bb0…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

80 FITS frames (53 MB), `HATP-3180403024212.FITS` → `HATP-3180403064512.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-3-180403.log (22ee9ca2c019…), FinalLightCurve_HAT-P-3 b_2018-04-02.png (210f86703ae2…), FinalLightCurve_HAT-P-3 b_2018-04-02.csv (6a7c94cef3e7…)

Compute resources

Wall time 125.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)